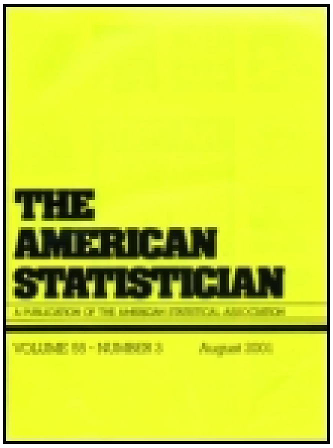

The American Statistician
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/utas20

AP Statistics: Building Bridges Between High School and College Statistics Education

Christine Franklin[a], Brad Hartlaub[a], Roxy Peck[a], Richard Scheaffer[a], David Thiel[a] & Katherine Tranbarger Freier[a]

[a] Christine Franklin is Senior Lecturer and Lothar Tresp Honoratus Honors Professor, Department of Statistics, University of Georgia, Athens, GA 30602. Brad Hartlaub is Professor, Department of Mathematics, Kenyon College, Gambier, OH 43022. Roxy Peck is Professor Emerita, Department of Statistics, California Polytechnic State University, San Luis Obispo, CA 93407. Richard Scheaffer is Professor Emeritus, Department of Statistics, University of Florida, Gainesville, FL 32603. David Thiel is Coordinator of K-12 Mathematics, Clark County School District, 3950 S. Pecos-McLeod, Las Vegas, NV 89121. Katherine Tranbarger Freier is a Senior Consultant in Biostatistics with Boston Biomedical Associates, 386 West Main Street, Suite 7, Northborough, MA 01532.
Published online: 24 Jan 2012.

Supplementary materials for this article are available online. Please click the TAS link at http://pubs.amstat.org.

AP Statistics: Building Bridges Between High School and College Statistics Education

Christine FRANKLIN, Brad HARTLAUB, Roxy PECK, Richard SCHEAFFER, David THIEL, and Katherine TRANBARGER FREIER

After providing a brief history of the AP Statistics program and a description of the AP Statistics course content, exam and grading, the paper presents a discussion of current challenges for statistics education in the schools and a look at opportunities for the statistics profession, especially college faculty, to aid the AP Statistics program so as to improve statistics teaching in both venues and thus strengthen the quantitative literacy of future generations of high school or college graduates. This article has supplementary material online.

KEY WORDS: Statistics education; Statistics standards; Teacher education.

1. INTRODUCTION

Advanced Placement (AP) Statistics is a course designed to provide high school students who are ready for college-level work the opportunity to experience an academically rigorous introduction to statistics and data analysis while still in high school. Not only does it provide an introduction to the discipline of statistics, it also invites school administrators to consider the role of statistics education in the lives of students, allows interested teachers to engage in an exciting "new" subject in the mathematical sciences, and opens doors for the statistics profession to converse with high school teachers and students on the important role of statistics in modern society.

AP Statistics provides a substantive fourth year of mathematics for the many students who will not take calculus in high school but have the opportunity to (or requirement of) taking four years of mathematics. The continued growth of the AP Statistics program presents both challenges and opportunities for the statistics profession. This article provides a brief history of the AP Statistics program, a description of the AP Statistics course curriculum and exam, a discussion of current challenges, and a look at how the statistics profession can help shape the future of the AP Statistics program to the benefit of future generations of citizens and the statistical profession.

Christine Franklin is Senior Lecturer and Lothar Tresp Honoratus Honors Professor, Department of Statistics, University of Georgia, Athens, GA 30602 (E-mail: *chris@stat.uga.edu*). Brad Hartlaub is Professor, Department of Mathematics, Kenyon College, Gambier, OH 43022 (E-mail: *hartlaub@kenyon.edu*). Roxy Peck is Professor Emerita, Department of Statistics, California Polytechnic State University, San Luis Obispo, CA 93407 (E-mail: *rpeck@calpoly.edu*). Richard Scheaffer is Professor Emeritus, Department of Statistics, University of Florida, Gainesville, FL 32603 (E-mail: *rls907@bellsouth.net*). David Thiel is Coordinator of K-12 Mathematics, Clark County School District, 3950 S. Pecos-McLeod, Las Vegas, NV 89121 (E-mail: *dthiel@interact.ccsd.net*). Katherine Tranbarger Freier is a Senior Consultant in Biostatistics with Boston Biomedical Associates, 386 West Main Street, Suite 7, Northborough, MA 01532 (E-mail: *kfreier@boston-biomedical.com*).

2. OVERVIEW OF AP STATISTICS

Advanced Placement courses are primarily first-year college level courses taught in high school. Upon completion of an AP course, a student seeking college credit can take a secure exam written by a test development committee comprised of both college and high school faculty. The AP Statistics exam consists of multiple-choice and free-response sections. Using rigorous rubrics that emphasize statistical thinking rather than just calculation skill, the free-response section is scored by experienced college faculty and high school statistics teachers. Many universities and colleges grant course credit and placement to students who score well on the exam.

AP Statistics was first taught in the 1996–1997 school year, with the first exam given in May of 1997. Approximately 7500 students took the first AP Statistics exam. Since that time, the number of students taking the exam has increased to well over 100,000 per year, with approximately 60% receiving an overall score of 3, 4, or 5 (see Table 1). A score of 3 is the lowest score for which a student might receive college credit; some colleges and universities require a 4 or a 5 for credit. The large number of students who are now being introduced to our discipline in high school presents both opportunities and challenges for our profession, some of which are addressed in the following.

2.1 History and Impact

Although the first AP Statistics courses were taught in the 1996–1997 academic school year, many years of work preceded this inaugural year. A number of statistics education initiatives in the 1980s, including the NSF-funded quantitative literacy programs managed by the American Statistical Association (ASA), were influential in shaping K-12 curriculum standards, which set the stage for AP Statistics. In 1989, the National Council of Teachers of Mathematics (NCTM) published its *Curriculum and Evaluation Standards* (NCTM 1989), which contained a curriculum strand in data analysis and probability, establishing a place for data analysis and probability in the K-12 curriculum alongside the more traditional mathematics content strands of number and operations, algebra, and geometry. This strand was strengthened in the revision of the standards

Table 1. Number of operational AP statistics exams and % scoring 3 or higher.

Year	Exams	% scoring 3 or higher
1997	7667	62.2
1998	15,488	59.7
1999	25,240	57.1
2000	34,118	53.7
2001	41,609	59.7
2002	49,824	56.8
2003	58,230	61.9
2004	65,878	59.9
2005	76,786	60.7
2006	88,237	60.2
2007	98,033	58.8
2008	108,284	59.3
2009	116,876	58.8
2010	129,899	58.7
2011	137,136	58.5

NOTE: Source: The College Board (*http://apcentral.collegeboard.com*).

document, *Principles and Standards for School Mathematics* (NCTM 2000), in 2000. Including data analysis in the K-12 curriculum promoted statistical thinking through the collection and analysis of real data, which caught the interest of many students and teachers, who found this content to be engaging, useful, intellectually challenging, and helpful to understanding applications of mathematics. The inclusion of statistics in the mathematics curriculum broadened the view of mathematical science in the schools. Unfortunately, early interest in teaching data analysis throughout the grade levels started to wane by the mid 1990s. Even with access to good curricular materials, teachers spent relatively less time on data analysis than recommended, compared to the other main components of the math curriculum (Tarr et al. 2006).

About this time, The College Board, which had been exploring possibilities for expanding its Advanced Placement course offerings in the mathematical sciences for many years, revisited its consideration of a statistics course. The arguments for such a course were now more persuasive and, after careful consideration, a decision was made to develop a course in introductory statistics. The success of the AP Statistics program in attracting the interest of mathematics teachers and students along with the support of school administrators and mathematics educators has been a major factor in the strong re-emergence of statistics in the K-12 standards of many states and in programs of national influence such as the Achieve Benchmarks (*http://www.achieve.org*), the *College Board Standards for College Success: Mathematics and Statistics* (College Board 2006), the mathematics framework of the National Assessment of Educational Progress (NAEP) (*http://nces.ed.gov/nationsreportcard/mathematics*), and the *Common Core State Standards for Mathematics* (National Governors Association and the Council of Chief State School Officers 2010) developed under the auspices of the National Governors Association and the Council of Chief State School Officers.

2.2 Course Content and Pedagogy

The AP Statistics course was designed as a modern introduction to the concepts of statistics and the methods of data analysis. The course developers were very forward-looking, and although the course was developed years before ASA published its *Guidelines for Assessment and Instruction in Statistics Education (GAISE)—College Report* (*http://www.amstat.org/education/gaise*), the content and pedagogy of the AP Statistics course is consistent with the recommendations of that report. For example, design of experiments, design of sample surveys, and simulation have been part of the AP Statistics course since the initial offering and no major content revisions are planned for the immediate future.

Even though the AP Statistics course is based on a one-semester introductory college course, it is usually taught in high school as a year-long course or in other scheduling patterns that provide more time than does the traditional three-hour college course. This allows teachers to develop students' statistical reasoning and to integrate active learning strategies throughout the course. The course is organized around four broad content areas: exploring data, sampling and experimentation, anticipating patterns (including probability and simulation), and statistical inference. A more detailed description of the course content can be found in the AP Statistics Course Description (*http://www.collegeboard.com/student/testing/ap/sub_stats.html?stats*).

2.3 The AP Statistics Exam

The AP Statistics exam is given in May of each year. The exam is made up of 40 multiple-choice questions and 6 free-response questions. One of the free-response questions is an investigative task, which is weighted more heavily than the other five in the scoring of the exam. The investigative task requires students to integrate concepts and methods from two or more content areas in a new setting. For example, the investigative task from the 2009 exam asked students to explain how the ratio of the sample mean to the sample median could be interpreted as a measure of skewness, to use an observed value of the ratio and a simulated sampling distribution to draw a conclusion about the shape of a population distribution, and to propose a different statistic for measuring skewness and explain how it measures skewness.

As an example of what might be covered in an entire free-response section, the free-response questions from 2002 asked students to interpret graphical displays, reason using interval estimates, show understanding of some basic principles of experimental design (including blocking, randomization, and blinding), use the normal distribution to answer questions about the long-run behavior of a random variable and of a linear combination of random variables, interpret a correlation coefficient, assess the impact of influential points in a regression analysis, construct and interpret a confidence interval, and demonstrate the ability to carry out a hypothesis test and to interpret the results in context. Free response questions from past exams can be found at *http://www.collegeboard.com/student/testing/ap/statistics/samp.html?stats*.

The free-response questions are scored holistically; each question is scored as either complete, substantial, developing, minimal, or incorrect. The rubrics that guide the scoring recognize that solutions have both a statistical knowledge component and a communication component. To receive the top score (complete response), the student must use appropriate methods to arrive at a correct answer and also communicate the results clearly and use correct terminology. A description of the expectations in terms of statistical knowledge and communication for each of the score levels can be found in the *AP Statistics Teacher's Guide* (page 212, *http://apcentral.collegeboard.com/apc/public/courses/teachers_corner/2151.html*).

With the current emphasis placed on high stakes testing nationally at the K-12 level, mathematics educators are faced with the challenge of how best to assess student learning of statistics. The AP Statistics Exam can serve as a model for how to assess students with multiple-choice and open-ended questions asking students to reason statistically (both conceptually and analyzing realistic data) instead of the more traditional questions that simply ask students to compute numerical summaries such as the mean and standard deviation. The College Board AP Statistics Test Development Committee has also provided a model for the collaborative development, training and use of scoring rubrics for open-ended problems. Professional development in the use of scoring rubrics occurs at both the AP Statistics Reading in June and throughout the year during AP Statistics workshops for teachers (Garfield and Franklin 2011).

2.4 The Reading—A Collaboration Between College Faculty and AP Statistics Teachers

The free-response section of the AP Statistics exam is scored during a one-week reading held in June. Graders are experienced college and high school teachers of statistics who come together in a productive and collegial collaboration. Readers are trained in how to apply the scoring rubrics and work together to score the exams. Each high school reader is partnered with a college reader and these teams consult regularly on questions about student responses. In addition, readers are organized into tables of 8 readers and each table has a table leader. Table leaders are experienced readers who answer questions and also back-read sample papers from each reader to ensure that the scoring rubrics are being correctly and consistently applied.

To someone who has not participated in an AP Statistics reading, the idea of spending an entire week grading exams probably sounds awful! But both college and high school readers come back year after year because of the opportunity the reading provides for collaboration and professional development. High school teachers of statistics, who may not have an extensive background in statistics, report that they learn a great deal from working with the scoring rubrics and the associated training provided, and from the opportunity to work and socialize alongside college faculty. Teachers remark that the reading is the "best professional development that an AP Statistics teacher can get." From the college reader perspective, there is much to be learned about teaching and creative classroom activities from colleagues teaching in high school. Talented and excited high school teachers of statistics have made substantial contributions to the pedagogy of modern statistics, including the use of active learning, uses of technology in the classroom, and formative assessments that enhance learning. The reading offers the opportunity to interact with and learn from some of the most outstanding and dedicated teachers of statistics at the school level. This provides a unique and professionally rewarding opportunity for statisticians to contribute expertise in the preparation of students to become statistically literate citizens and possibly future statisticians.

3. CURRENT CHALLENGES

With statistics and data analysis now a more visible part of the K-12 curriculum in many states and with increasing enrollments in AP Statistics, many students are introduced to statistics prior to college. And, for the more than 100,000 students per year in AP Statistics, the first college-level introduction to statistics is also occurring in the high school. This presents a number of challenges for the statistics profession, particularly with respect to curriculum alignment, teacher preparation, student recruitment, and the development of appropriate subsequent courses for students not majoring in statistics that take advantage of an AP Statistics background.

3.1 Curriculum Alignment

In AP subjects, great care is taken to ensure that the content of the AP course closely matches the content of the corresponding college course. This presented an interesting challenge for the developers of the original AP Statistics course description. Roberts, Scheaffer, and Watkins (1999) noted, in an article in *The American Statistician*, some of the difficulties encountered, including the fact that there was not a consensus on the course content at the college level and that the introductory statistics course was offered by many different departments at some universities. Fortunately, the developers were very insightful, and they anticipated many of the changes in the content and pedagogy of the introductory statistics course that have occurred over the last decade. Rather than matching the fairly traditional introductory statistics course that was the norm in the 1990s, the developers took the somewhat risky path of defining a modern course that focuses on conceptual understanding, integrated the use of technology both for data analysis and as a tool for developing conceptual understanding, and included coverage of methods of data collection. Also, to foster acceptability of the AP Statistics course by colleges and universities, the AP Statistics course content covers more content topics than most students will see in a college introductory course. Several AP Statistics topics are typically covered in the second college statistics course.

Thus, at the time AP Statistics was first taught, it did not "match" a typical college introductory statistics course. But many who took a careful look at the AP course description and philosophy, including the authors of this article, found it to be a richer and more effective introduction to the discipline. In the years since the introduction of AP Statistics, the first course in statistics at many colleges has been transformed in response to research in statistics education about how students learn, *GAISE College Report* recommendations that the course focus on conceptual understanding, and the model provided by the AP Statistics course. The curriculum challenge will now be to learn from both the AP Statistics course and the reforms taking place at the college level in order to improve both courses and keep them in alignment.

An added curriculum challenge for both AP Statistics and the comparable college course is presented by the changing K-12 standards for mathematics. The integration of statistics and data analysis into the K-12 curriculum will have major implications for both courses. For example, several states have used the *GAISE Pre K-12 Framework Report* (Franklin et al. 2007) to inform the revision of state standards. Georgia provides one example of this. A visit to the website, *https://www.georgiastandards.org/Standards/Pages/BrowseStandards/BrowseGPS.aspx* shows significant integration of statistics into the math curriculum, especially at grades 6–12. Other examples of substantial integration of statistics at K-12 can be found in Colorado (*www.cde.state.co.us/cdeassess/documents/OSA/standards/math.htm#standards*), Wisconsin (*dpi.state.wi.us/standards/matintro.html*) and Ohio (*http://www.ode.state.oh.us/GD/Templates/Pages/ODE/ODEDetail.aspx?Page=3&TopicRelationID=1704&Content=86689*). The GAISE Framework has also become an instrumental document in defining the role of statistics within the school mathematics curriculum in national documents such as NCTM's *Reasoning and Sense-Making: Focus on High School Mathematics* (2009), and the documents referenced at the end of Section 2.1. AP Statistics has also helped to rekindle the notion that statistics at some level should be a necessary component of the K-12 curriculum, not just a single course in high school for select students. If this integration becomes a reality, the very nature of the introductory college-level statistics course (which includes AP Statistics) will need to change. Another challenge for many institutions will be the development and refinement of the second course in statistics, which is addressed later in this section.

3.2 Teacher Preparation

Availability of high school teachers qualified to teach statistics was a concern even in early discussions about the AP Statistics program. This initial concern subsided when capable, serious, and energetic teachers excited about teaching this "new" mathematical sciences course seemed to come out of the woodwork at many high schools. These teachers, along with a core group of activist college teachers of statistics, created a network that resulted in hundreds of workshops on teaching statistics, the development of strong teaching materials, and a collegial environment for sharing of ideas. The electronic discussion group for AP Statistics is very active, with more than 3500 AP Statistics teachers, college and university professors, and professional statisticians contributing to the discussion. For many teachers, this professional interaction has been invaluable.

However, as the number of students taking AP Statistics and the number of schools offering the course has continued to increase, the number of capable teachers has not kept pace. Continued and rapid growth has resulted in a great need for high school mathematics teachers who are adequately prepared to teach statistics. The College Board has made the professional development of AP Statistics teachers one of its highest priorities. In the summer of 2009, 557 teachers attended week-long AP Statistics summer institutes. Hundreds more teachers attended one- and two-day workshops during the 2009–2010 school year. Not only is there a need to address the preparation of teachers early in their careers, there is increasing recognition that many of the current master teachers in AP Statistics are nearing retirement. This has implications for both preservice and in-service teacher education.

Efforts to enhance the skills of both current and future AP Statistics teachers lag behind the need, and colleges of education must be encouraged to take up the challenge of preparing more teachers for this curriculum. Such enhanced teacher education would pay dividends not only for the AP Statistics program but also in the wide variety of curricular settings across K-12 mathematics in which data analysis and probability are destined to play much stronger roles. For example, collaboration between statisticians and mathematics educators at the University of Georgia has resulted in a statistics course specifically for preservice high school teachers. The course has been successfully taught since 1998. This collaboration has also led to the development of statistics courses for middle school teachers and elementary school teachers (Franklin and Mewborn 2006). However, feedback from College Board consultants and AP Statistics teachers indicates that there is a considerable need for more preservice training, especially in the areas of statistical content and active learning. Although workshops are beneficial for professional development of teachers, much work remains to be done to integrate statistics as a major component of undergraduate and graduate degree programs for teachers.

3.3 A Next Course

A large number of students are now coming to the university having had previous experience with statistics in high school, and this number is destined to increase. This presents the opportunity to explore new ideas in both content and pedagogy within the context of the traditional introductory statistics course, and it also provides a potential sizable audience for a second course in statistics.

Colleges and universities need to respond by rethinking the "traditional" college-level introductory course and providing an appropriate second course for students who earned high scores on the AP Statistics exam. Many students with great academic ability and interest in quantitative areas are not able to further their knowledge of statistics within mathematics or statistics departments because of the lack of appropriate course offerings. Colleges and universities should pay more attention to AP Statistics students entering their institutions, providing credit for high scores on the exam and steering these students toward other statistics courses that will maintain their interest and enhance their knowledge of the subject. One possibility is a second course that is built around statistical modeling, with an applied emphasis on regression, analysis of variance, and modeling categorical data. Reports favoring this view can be seen in ASA's Undergraduate Statistics Education Initiative of 2000 (*http://www.amstat.org/meetings/JSM/2000/symposium.html*).

College and university faculty members in statistics must recognize the potential of the AP Statistics students entering their institutions and should look after these students with greater care. Such students should neither be forced to retake an introductory course, nor be sent off without guidance as to what they should do next to build upon their interest in statistics. If there is no appropriate course for such students, one should be developed.

Universities with strong undergraduate majors in statistics, among them California Polytechnic State University (Cal Poly),

Table 2. AP Statistics credit at University of Georgia.

Date	Score 3	Score 4	Score 5	Total
Fall 2001	108	103	40	251
Fall 2002	107	95	35	237
Fall 2003	204	124	64	392
Fall 2004	173	144	53	370
Fall 2005	190	182	70	442
Fall 2006	259	198	85	542
Fall 2007	276	236	79	591
Fall 2008	274	270	73	617
Fall 2009	306	279	103	688
Fall 2010	337	297	91	725
Total	2234	1928	693	4855

NOTE: Source: University of Georgia Registrar's Office.

Brigham Young University, University of Georgia (UGA), and North Carolina State, report increased numbers of students choosing statistics as an undergraduate major or minor, and give the AP Statistics program much of the credit for this because many of the students choosing statistics as a major report that AP Statistics sparked their interest in the field. For example, UGA reports that from Fall 2001 through Fall 2010, 4855 students were granted credit for the introductory statistics course based on a score of 3 or higher received on the AP Statistics exam (see Table 2). From Fall 2002 through Fall 2010, the total number of undergraduate statistics majors graduating was 89, a noticeable increase from the years previous to 2002. The strong academic backgrounds of the recent graduates as compared to the graduates of the years prior to 2002 are also noteworthy. The UGA department attributes much of this growth to AP Statistics. Currently, the department averages 50 majors and 30 minors. Also noteworthy are the numbers of UGA undergraduates who now choose to pursue advanced degrees in statistics or biostatistics. At Cal Poly, where students must apply for admission to a specific undergraduate major, the number of students applying to the statistics major has more than tripled since the AP Statistics program began. The Statistics Department now receives more applications than it can accommodate, and the number of students majoring in statistics has grown to about 75. However, the increase in undergraduate majors may not be occurring uniformly at colleges nationwide. According to the CBMS survey of undergraduate enrollment (CBMS 2007), there was not a noticeable increase in bachelor degrees awarded in statistics by 2005, as that number remained stable at a little over 500 for both 2000 and 2005.

Very few small liberal arts colleges have a major in statistics, so they would not be included in the CBMS survey results. However, many schools are experiencing larger enrollments in statistics courses. For example, over the past 10 years Kenyon College has increased the number of sections of introductory statistics from four per year to six per year. New faculty members have been hired to cover the increased demand for these courses, but budget constraints will not allow additional expansion of the faculty. Bowdoin College and Oberlin College have been unable to meet the current demand for introductory statistics courses.

3.4 What Happens to AP Statistics Students?

Little is known about what happens to AP Statistics students once they reach college. One exception is Katherine Tranbarger Freier, who is profiled in this article. To place the more than 60,000 students each year now scoring 3 or better on the AP Statistics exam into perspective, Table 3 shows enrollment figures from the CBMS survey (CBMS 2007) for each half decade since 1990. From 2000 to 2005 enrollment increases in introductory statistics were 0% in statistics departments, 8.8% in mathematics departments, and 58% in two-year colleges. Since four-year college enrollments were up 14% over that period (two-year college enrollments were up 12%) this shows a relatively slower growth of introductory statistics students at four-year institutions. AP Statistics might have contributed a bit to this slower growth, but the major contributor is the jump in enrollments within the two-year colleges. More importantly, the enrollments in upper level courses have remained fairly flat over a decade or more. While some AP Statistics students are choosing to major in statistics, it appears that colleges and universities are missing an opportunity to take advantage of a strong AP Statistics program by developing new courses beyond the introductory level to attract these students and allow them to further their statistics education.

College and university faculty currently teaching statistics can help improve the AP Statistics knowledge base in several potentially important ways. Because credit for AP Statistics is often granted by an admissions administrator, statistics faculty may not have any knowledge of these students. Statistics faculty should develop a method to receive information on these students and to inform them of further statistics opportunities on campus. Second, college faculty should build contacts with local high school AP Statistics teachers to (gently) offer their assistance with the course and to learn how those doing well on the exam have fared at the college level. Third, statisticians might offer their professional services to the College Board, CBMS, ASA, and other groups interested in high school-to-college articulation to help design surveys for getting good data on what happens to AP Statistics students at the college level

Table 3. Enrollments in statistics courses, fall terms (thousands).

	Mathematics departments				Statistics departments				Two-year college math			
	1990	1995	2000	2005	1990	1995	2000	2005	1990	1995	2000	2005
Intro	87	115	136	148	30	49	54	54	54	72	74	117
Upper	38	28	35	34	14	16	20	24	0	0	0	0
Total	125	143	171	182	44	65	74	78	54	72	74	117

NOTE: Source: CBMS Survey.

and beyond. What are their college majors? Are they adequately prepared for further statistics courses? Are they adequately prepared for other courses that require introductory statistics as a prerequisite? If not prepared, what changes should be made in the AP Statistics curriculum? Such efforts could be of immeasurable help in building a quantitatively literate society and strengthening the statistics profession.

4. LOOKING TO THE FUTURE

In spite of the challenges discussed in the previous section, many opportunities lie ahead and the statistics profession can play a major role in strengthening AP Statistics and shaping its future in a way that will benefit both the profession and society at large. As a profession, we have long argued that some knowledge of statistics is essential for effective citizenship—everyone should be able to evaluate arguments based on data and make sound decisions informed by data. Only about half of all high school graduates go on to college, so if we really believe this we must become advocates for statistics education in K-12. In addition, we must become activists to ensure a quality statistics curriculum in K-12 and AP Statistics and to ensure that current and future teachers at all levels are prepared to teach statistics.

To garner the support of mathematicians, mathematics educators, and others who think deeply about the school curriculum, arguments based on skills for work, college, or citizenship must be bolstered by arguments based on the intellectual merit of statistical thinking. It must be made clear to all concerned that statistical reasoning is essentially the reasoning of discovery. George Polya, a famous mathematician, clearly described this reasoning—the inferential reasoning of science and everyday life by which *new knowledge is obtained*—as an important part of mathematical reasoning that should be included in the school mathematics curriculum (Polya 1959).

Moving in this direction will require greater involvement of professional statisticians. The intellectual merit, not to mention the useful life skills, that come from studying statistics must be made clear to the various education constituencies that control school curricula at the state and local levels, including parents, teachers, school administrators, school board members, and politicians. This cannot be accomplished nearly as effectively by a statistics teacher as it can by a respected member of the research or business community, for example, who can speak knowledgeably and passionately about economic, medical, biological, social, or other applications of the subject. Research mathematicians have taken mathematics education at the school level seriously for the past decade or more; now is the time for research statisticians to support statistics education.

Faculty members in statistics at institutions with teacher education programs should make it a point to visit that area of campus to learn what is happening with the statistics education of future mathematics teachers. Chances are the initial effort could produce some fruitful exchanges between statisticians and teacher educators, who may surprisingly welcome the exchanges if they are handled with respect and appropriate academic integrity. Many mathematics educators are advocates for more data analysis in the K-12 curriculum and welcome collaboration with statisticians.

A great opportunity also exists for recruiting strong students to our discipline. Institutions with undergraduate statistics majors often recruit capable students in introductory statistics courses into statistics majors and minors. We need to think carefully about how we can most effectively recruit talented AP Statistics students and bring them into the fold.

There is great potential for expanding and improving K-12 statistics education and for supporting continued growth of AP Statistics—as a profession we must explore ways to make the most of this potential. Building a stronger bridge between high schools and colleges can be a fruitful beginning.

SUPPLEMENTARY MATERIALS

AP Profile: Read about a former AP Statistics student, Kate Freier, who chose statistics as her major in college and eventually career field. (Supplemental - An AP Profile.pdf)

[Received May 2009. Revised July 2011.]

REFERENCES

College Board (2006), *College Board Standards for College Success: Mathematics and Statistics*, New York: Author. Available at *http://professionals.collegeboard.com/k-12/standards*. [178]

Conference Board of the Mathematical Sciences (2007), *Statistical Abstract of Undergraduate Programs in the Mathematical Sciences, Fall 2005*, Providence, RI: American Mathematical Society. [181]

Franklin, C., and Mewborn, D. (2006), "The Statistical Education of Grades Pre-K-12 Teachers," in *Thinking and Reasoning With Data and Chance: 68th Yearbook*, Reston, VA: National Council of Teachers of Mathematics, pp. 335–344. [180]

Franklin, C., Kader, G., Mewborn, D., Moreno, J., Peck, R., Perry, M., and Scheaffer, R. (2007), *Guidelines and Assessment for Instruction in Statistics Education (GAISE) Report: A Pre-K-12 Curriculum Framework*, Alexandria, VA: American Statistical Association. [180]

Garfield, J., and Franklin, C. (2011), "Assessment of Learning for Learning and as Learning in Statistics Education," in *Teaching Statistics in School Mathematics—Challenges for Teaching and Teacher Education: A Joint ICMI/IASE Study*, New York: Springer-Verlag. [179]

National Council of Teachers of Mathematics (NCTM) (1989), *Curriculum and Evaluation Standards*, Reston, VA: Author. [177]

——— (2000), *Principles and Standards for School Mathematics*, Reston, VA: Author. [178]

——— (2009), *Focus in High School Mathematics: Reasoning and Sense-Making*, Reston, VA: Author. [180]

National Governors Association and the Council of Chief State School Officers (2010), *Common Core State Standards*, Washington, DC: Authors. Available at *http://corestandards.org/*. [178]

Polya, G. (1959), "Mathematics as a Subject for Learning Plausible Reasoning," *Mathematics Teacher*, 52, 7–9. [182]

Roberts, R., Scheaffer, R., and Watkins, A. (1999). "Advanced Placement Statistics—Past, Present, and Future," *The American Statistician*, 53, 307–320. [179]

Tarr, J. E., Chávez, O., Reys, R. E., and Reys, B. J. (2006), "From the Written to the Enacted Curricula: The Intermediary Role of Middle School Mathematics Teachers in Shaping Students' Opportunity to Learn," *School Science and Mathematics*, 106, 191–201. [178]